Oluwatosin Ayeni

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H Company

Dear Hiring Team,

I am writing to apply for the Research Engineer, Model Inference & Serving role at H Company. The challenge of owning the end-to-end inference stack for multimodal agentic models — optimising latency, throughput, and cost simultaneously — is exactly the problem class I have been building toward.

Three specific achievements directly relevant to this role:

- **End-to-end inference stack ownership:** I benchmarked and tuned vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2 — measuring TTFT, TPOT, and throughput across concurrency 1–128. RadixAttention prefix caching: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 5,965 tok/s peak throughput; 100% SLO compliance. Complete inference stack ownership from hardware selection through kernel profiling to live production dashboard. Live: inference-web-theta.vercel.app/benchmarks/llm
- **Multimodal inference characterisation:** I built *lab-vllm-omni*, a phased inference lab for vLLM-Omni covering all 5 output modalities — text, audio, image, video, structured output. Characterised throughput, memory pressure, and scheduling behaviour across modalities, directly informing the serving challenges for multimodal agentic systems.
- **Production cost and latency reduction at scale:** Via vLLM/SGLang optimisation on RTX 4090, I achieved 28× LLM cost reduction and 3.8× throughput improvement. I also built the *inference-optimizer* production gateway delivering 3.87× throughput, 61.4% cost reduction, and 64.3% cheaper routing across four providers — the multi-provider routing pattern that underpins agentic inference at scale.

H Company's focus on building frontier multimodal agentic models — and the inference serving challenges that come with them — is where I want to contribute. I'm happy to be in the London office 3 days/week and am open to relocation.

Yours sincerely,

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